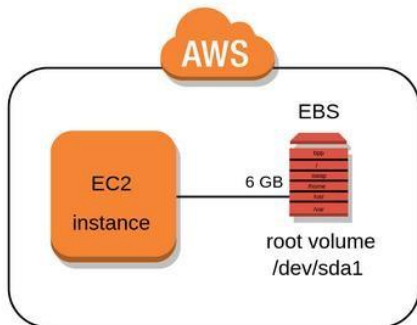


# Lab: Attach an EBS volume to your EC2 instance

## Overview

Amazon Elastic Block Store (Amazon EBS) offers persistent storage for Amazon EC2 instances. Amazon EBS volumes are network-attached and persist independently from the life of an instance. In this lab, you create and attach an EBS volume to your EC2 instance.



## Task 1: Create Volume

Open the **EC2 Management Console** browser tab.

1. In the left navigation pane, under **Instances**, choose **Instances**.

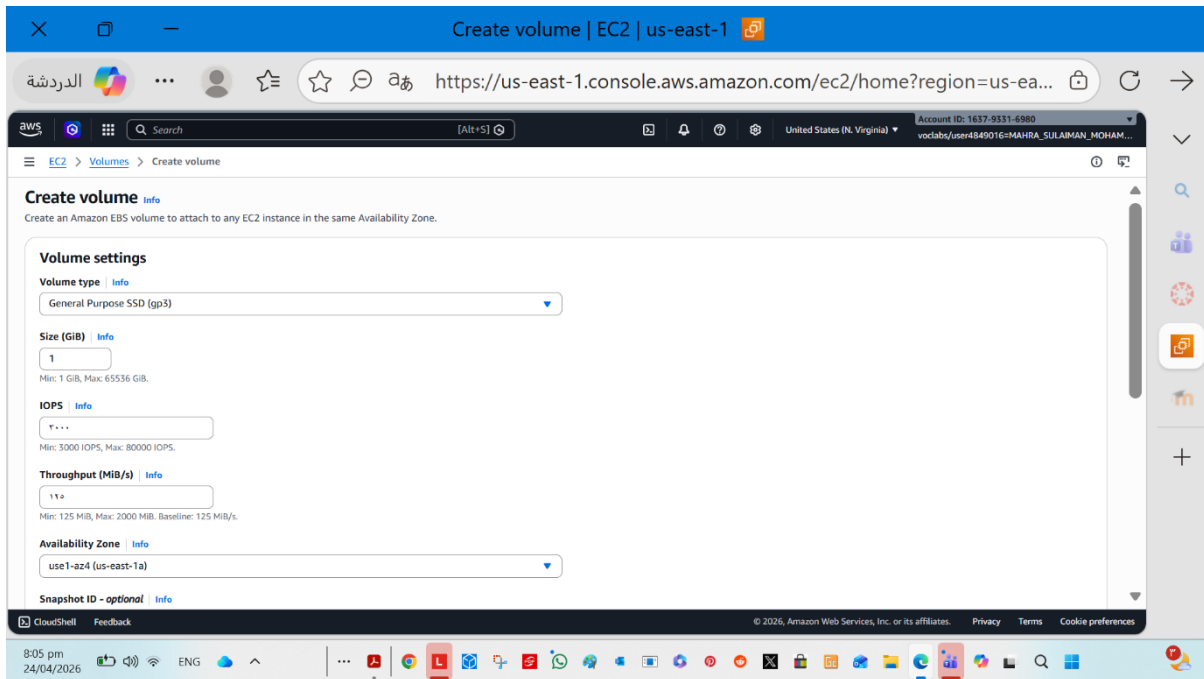
Note: If no instance is present do Lab 1 and create a Linux Web Server instance.

2. Select the **Web Server** instance, and in the **Networking** tab below, note the **Availability Zone** in which your instance is running. (see the alphabet a, b, c). Note: The EBS volume you will soon create will need to be in the same Availability Zone.

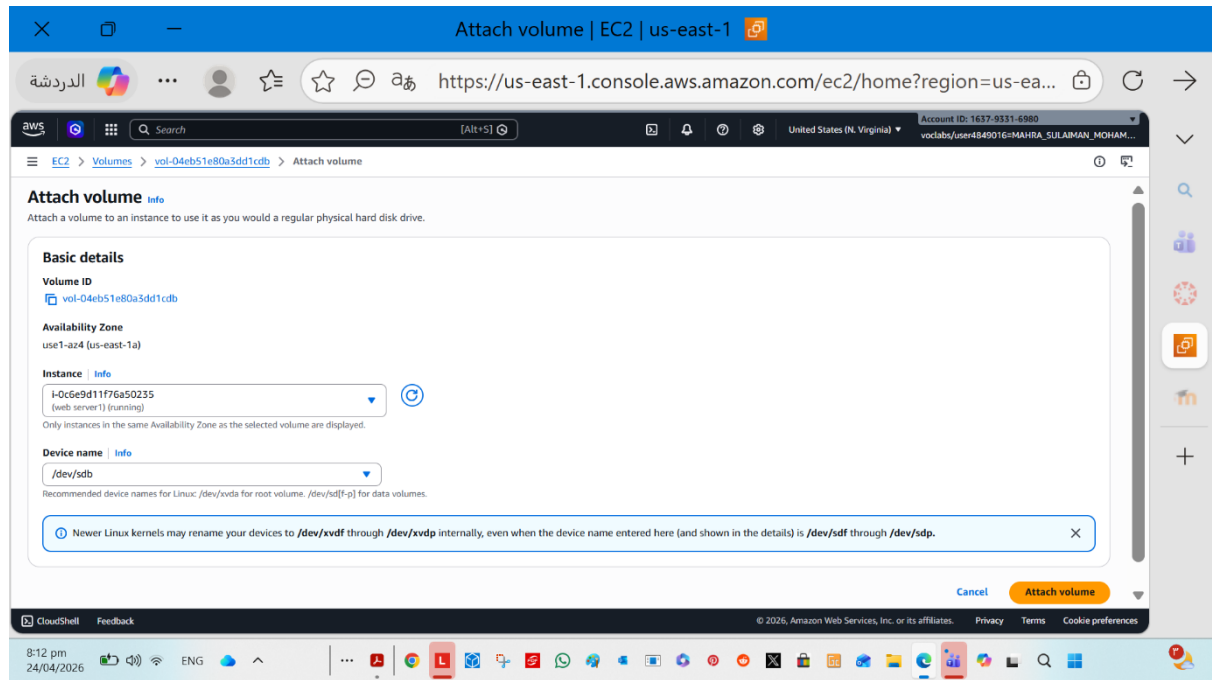
| Name             | Instance ID         | Instance state | Instance type | Status check      | Alarm status | Availability Zone | Public IPv4 DNS      |
|------------------|---------------------|----------------|---------------|-------------------|--------------|-------------------|----------------------|
| Web server Ma... | i-07a869240443fe288 | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1c        | ec2-3-239-188-63.co  |
| web mahra        | i-0b55b4ab36be9d351 | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1c        | ec2-3-234-220-185.co |
|                  | i-0bf3fadc31b201515 | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1c        | ec2-44-200-34-65.co  |
|                  | i-06d971ab30085d52d | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1c        | ec2-44-204-247-119.  |
| MyWindowsSe...   | i-0652efe1e8aa04166 | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1f        | ec2-44-200-139-3.co  |
| web server1      | i-0c6e9d11f76a50235 | Running        | t2.micro      | 2/2 checks passed | View alarms  | us-east-1a        | ec2-3-83-231-174.co  |
| windows server   | i-09651f829b6af49ca | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1f        | ec2-3-236-210-96.co  |
|                  | i-0f77fb4df36872b42 | Running        | t3.micro      | 3/3 checks passed | View alarms  | us-east-1f        | ec2-44-213-64-214.co |

- 3.

4. In the left navigation pane, under **Elastic Block Store**, select **Volumes**.
  5. Select **Create volume**.
  6. For **Volume Type**: General Purpose SSD (gp2)
  7. For **Size**, enter 1 to create a volume with 1 GiB.
  8. For **Availability Zone**, select the same Availability Zone that your EC2 instance is running in.
  9. Scroll down and select **Create volume**.
- The new volume appears in the volumes list with a state of **available**.



10. Select the new 1 GiB size volume. Then, choose **Actions**, and **Attach volume**.
  11. Select the **Instance** drop-down menu, and then select your EC2 instance. The list of instances will automatically populate.
  12. Select the **Device Name** drop-down menu, and then select a device name **/dev/xvdb**. (you can choose any available device name and note down to use it later.)
  13. Select **Attach volume**.
- The state of the volume changes to *in-use*. The new volume is now attached to your EC2 instance.



## Task 2: Mount new disk on Linux instance

The `df` (disk free) command is used to display information about the disk space usage on file systems. It provides details such as the total, used, and available disk space for each mounted file system. Run the following command on your instance to view its mounted volumes

**\$ `df -h`**

you won't see the volume yet because we need to create file system on disk or mount the volume.

The `lsblk` (list block devices) command in Linux is used to display information about all available block devices. The following command, however, can show all block devices attached to the instance.

**\$ `lsblk`**

To make a volume available, first identify the device to mount. In the previous procedure, the new volume was attached to **`/dev/xvdb`**

You need to format the volume before you can mount it.

`mkfs` (make file system) command format and create an `ext4` file system on the volume. Substitute the device name you used if you did not use `/dev/xvdb` when you attached the volume.

**\$ `sudo mkfs -t ext4 /dev/xvdb`**

To mount the device as `/mnt/newdisk`, create a new folder and mount new disk by running the following commands.

**\$ `sudo mkdir /mnt/newdisk`**

**\$ `sudo mount /dev/xvdb /mnt/newdisk`**

Now when you run the **df -h** command, you'll see new disk in output.

To view the contents of the new volume, run the following command.

## \$ ls /mnt/newdisk

Create a file in new disk and run again above command to see the file.

## \$ sudo nano /mnt/newdisk/gaurav.txt

The image displays two screenshots of an AWS EC2 Instance Connect terminal session. The top screenshot shows the initial state where a new EBS volume is being created and attached to the instance. The user runs `df -h` and sees the new volume `/dev/xvdb` listed. The bottom screenshot shows the user running `ls /mnt/newdisk` and seeing the file `gaurav.txt` created in the new volume.

**Top Screenshot:**

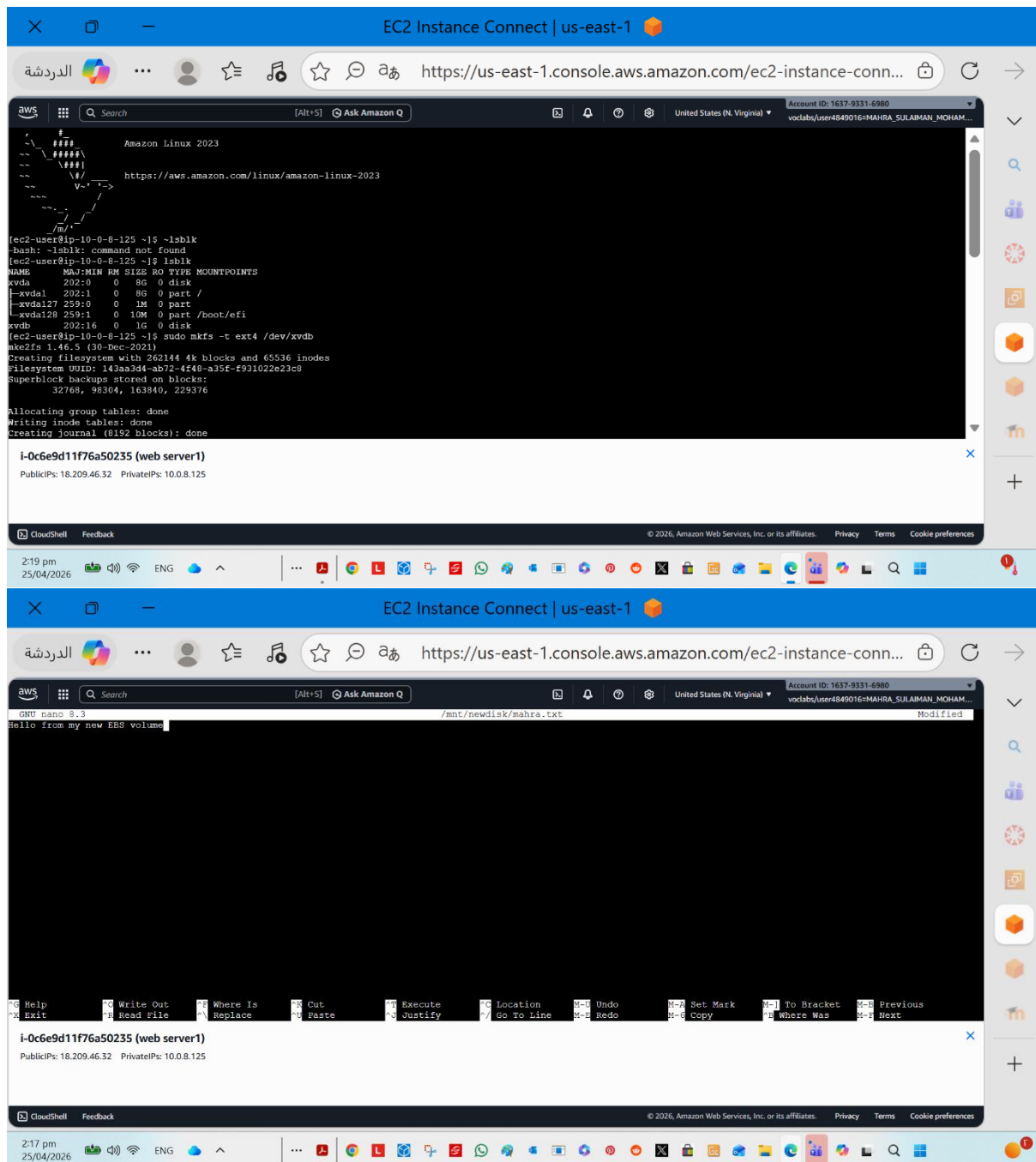
```
aws
Search [Alt+S] Ask Amazon Q
United States (N. Virginia) Account ID: 1637-9331-6980
voclabs/user484901G=MAHRA_SULAIMAN_MOHAM...

/dev/xvda1 8.0G 1.7G 6.3G 21% /
tmpfs 481M 0 481M 0% /tmp
/dev/xvda128 10M 1.3M 8.7M 13% /boot/efi
tmpfs 97M 0 97M 0% /run/user/1000
[ec2-user@ip-10-0-8-125 ~]$ sudo mkdir -p /mnt/newdisk
[ec2-user@ip-10-0-8-125 ~]$ sudo mount /dev/mnt/newdisk
mount: /dev/mnt/newdisk: can't find in /etc/fstab.
[ec2-user@ip-10-0-8-125 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0  4.0M   0% /dev
tmpfs           481M  0  481M   0% /dev/shm
tmpfs          193M 444K 192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  21% /
tmpfs           481M  0  481M   0% /tmp
/dev/xvda128    10M  1.3M  8.7M  13% /boot/efi
tmpfs           97M  0  97M   0% /run/user/1000
[ec2-user@ip-10-0-8-125 ~]$ sudo mount /dev/xvdb /mnt/newdisk
[ec2-user@ip-10-0-8-125 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0  4.0M   0% /dev
tmpfs           481M  0  481M   0% /dev/shm
tmpfs          193M 444K 192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  21% /
tmpfs           481M  0  481M   0% /tmp
/dev/xvda128    10M  1.3M  8.7M  13% /boot/efi
tmpfs           97M  0  97M   0% /run/user/1000
/dev/xvdb       974M  24K  907M   1% /mnt/newdisk
[ec2-user@ip-10-0-8-125 ~]$ sudo nano /mnt/newdisk/mahra.txt
[ec2-user@ip-10-0-8-125 ~]$
```

**Bottom Screenshot:**

```
aws
Search [Alt+S] Ask Amazon Q
United States (N. Virginia) Account ID: 1637-9331-6980
voclabs/user484901G=MAHRA_SULAIMAN_MOHAM...

Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
[ec2-user@ip-10-0-8-125 ~]$ sudo mkdir /mnt/newdisk
[ec2-user@ip-10-0-8-125 ~]$ sudo mount /dev/xdev /mnt/newdisk
mount: /mnt/newdisk: special device /dev/xdev does not exist.
[ec2-user@ip-10-0-8-125 ~]$ df -h
-bash: df-h: command not found
[ec2-user@ip-10-0-8-125 ~]$ sudo mount / dev/xvdb/mnt/newdisk
mount: dev/xvdb/mnt/newdisk: mount point does not exist.
[ec2-user@ip-10-0-8-125 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0  4.0M   0% /dev
tmpfs           481M  0  481M   0% /dev/shm
tmpfs          193M 444K 192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  21% /
tmpfs           481M  0  481M   0% /tmp
/dev/xvda128    10M  1.3M  8.7M  13% /boot/efi
tmpfs           97M  0  97M   0% /run/user/1000
[ec2-user@ip-10-0-8-125 ~]$ sudo mkdir -p /mnt/newdisk
[ec2-user@ip-10-0-8-125 ~]$ sudo mount /dev/mnt/newdisk
mount: /dev/mnt/newdisk: can't find in /etc/fstab.
[ec2-user@ip-10-0-8-125 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0  4.0M   0% /dev
tmpfs           481M  0  481M   0% /dev/shm
tmpfs          193M 444K 192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  21% /
tmpfs           481M  0  481M   0% /tmp
```



Lab compile by Dr. Gaurav Bhatia, Program Lead – Cloud Computing Specialization, UTAS, Oman.

